Sidon Subsets and Pair-Sum Multiplicities of Distinct Multinomial Coefficients

Felix Huber
Independent Researcher, Lucerne, Switzerland
felix.68@gmx.ch
ORCID: 0009-0005-1568-1579

Abstract

For a positive integer n, let the nth multinomial row be the set of distinct multinomial coefficient values arising from the partitions of n. We study two complementary questions about its additive structure: how large a Sidon subset can be, and how often pair sums collide in the full row. A product embedding between multinomial rows is the common mechanism. General Sidon extraction and prime-partition estimates give a superpolynomial lower bound for the largest Sidon subset. Long arithmetic progressions, replicated at separated scales through the same embedding, force a large Sidon defect and quantitative lower bounds for pair-sum multiplicities. We also obtain a product-growth upper constraint, an additive-energy bound, variable-number stabilization, and an explicit family of trinomial collisions.

Keywords: additive combinatorics; Sidon set; multinomial coefficient; pair sum; representation function; additive energy; integer partition; prime partition.
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1 Introduction

Multinomial coefficients are defined by products of factorials, but the set of distinct values in a fixed row can have surprisingly visible additive structure. For $n \geq 1$, let

$$\mathcal{M}_n = \left\{ \frac{n!}{p_1! \cdots p_t!} : p_1 + \cdots + p_t = n, \quad p_1 \geq \cdots \geq p_t \geq 1 \right\}$$

be the set of distinct multinomial coefficient values associated with the integer partitions of n. Already at $n = 5$,

$$\mathcal{M}_5 = \{1, 5, 10, 20, 30, 60, 120\}$$

contains the nontrivial relation

$$10 + 30 = 20 + 20.$$

At the same time, deleting just one term is enough to recover a set with unique pair sums: for example,

$$\{1, 5, 20, 30, 60, 120\}$$

is Sidon. By $n = 9$ the phenomenon is richer still: the sum 1764 has three distinct representations,

$$84 + 1680 = 252 + 1512 = 504 + 1260.$$

These small examples lead to two complementary questions. How much of $\mathcal{M}_n$ can be retained if every nontrivial pair-sum collision is forbidden? And what does the collision pattern look like when the entire set is kept?

A finite set S of integers is a *Sidon set* if every equality

$$x + y = u + v, \qquad x, y, u, v \in S,$$

is trivial, meaning that the unordered pairs $\{x, y\}$ and $\{u, v\}$ coincide. Sidon sets are classical objects of additive combinatorics; see O'Bryant's survey [7]. We write

$$s(n) = \max\{|S| : S \subseteq \mathcal{M}_n,\ S \text{ is Sidon}\}.$$

For the full set, let

$$r_n(t) = \#\{(x, y) : x, y \in \mathcal{M}_n,\ x \leq y,\ x + y = t\}$$

be the number of unordered representations of t, and put

$$C(n) = \sum_t \max(r_n(t) - 1, 0), \qquad D(n) = \#\{t : r_n(t) \geq 2\},$$
$$\mu(n) = \max_t r_n(t), \qquad T(n, k) = \#\{t : r_n(t) \geq k\} \qquad (k \geq 2).$$

Thus $s(n)$ measures how large a collision-free part can be, whereas $C(n)$, $D(n)$, $\mu(n)$, and $T(n, k)$ describe how collisions accumulate in the full set. The Sidon defect $M(n) - s(n)$, where $M(n) = |\mathcal{M}_n|$, is exactly the minimum number of elements that must be removed to eliminate every repeated unordered pair sum. Pair-sum representation functions are standard in additive number theory; see Nathanson [6].

The reason these two questions can be treated together is an elementary product embedding. Whenever $n_1 + \cdots + n_\ell = n$,

$$\binom{n}{n_1, \ldots, n_\ell} \mathcal{M}_{n_1} \cdots \mathcal{M}_{n_\ell} \subseteq \mathcal{M}_n.$$

Concatenating the partitions that represent elements of the smaller rows proves the inclusion. What matters is what the inclusion lets us do: an additive configuration found at a lower order can be transported to higher orders, multiplied by new factors, and placed at a scale separated from other configurations.

This one mechanism has two opposite uses. To build large Sidon sets, we place collision-free seed sets at multiplicative scales so far apart that no new cross-scale equality can occur. To force collisions, we first manufacture a long arithmetic progression inside one multinomial row and then place many separated copies of that progression inside a larger row. The element intervals of the copies are disjoint, so every Sidon subset must lose elements independently in every copy. Their pair-sum intervals are also disjoint, so the collisions produced inside different copies contribute independently to the representation profile. In this sense the collision-free and collision-rich halves of the paper are not separate problems: they are two readings of the same product structure.

There is also a second source of large Sidon subsets that does not use the separated-copy construction. Andrews, Knopfmacher, and Zimmermann studied $M(n)$ and showed, in particular, that $\mathcal{M}_n$ contains at least as many distinct values as there are partitions of n into primes [1]. General Sidon-extraction results then turn this abundance of values into a strong existence theorem. Abbott proved that every sufficiently large m-element set of integers contains a Sidon subset of size $\gg \sqrt{m}$ [2]; Bailleul and Riblet recently sharpened the constant [3]. Combining these facts gives our strongest general lower bound for $s(n)$. Here and below, log denotes the natural logarithm.

Theorem 1.1 (Sidon growth)**.** *We have*

$$\liminf_{n\to\infty} \frac{\log s(n)}{\sqrt{n/\log n}} \geq \frac{\pi}{\sqrt{3}}.$$

Equivalently, for every $\varepsilon > 0$,

$$s(n) \geq \exp\left(\left(\frac{\pi}{\sqrt{3}} - \varepsilon\right)\sqrt{\frac{n}{\log n}}\right)$$

for all sufficiently large n. In particular, $s(n)$ grows faster than every fixed power of n.

The product embedding gives a different kind of information: not just existence, but explicit configurations whose additive behavior can be followed from one order to the next. The key collision-rich configuration is a long arithmetic progression. If its length is q, the order needed to create it is asymptotic to $q^2/(2\log q)$. This makes $q \asymp \sqrt{n\log n}$ the natural scale at order n. At that scale, separated progression copies force the following defect estimate.

Theorem 1.2 (Sidon defect)**.** *With $M(n) = |\mathcal{M}_n|$,*

$$\liminf_{n\to\infty} \frac{M(n) - s(n)}{n^{3/2}\sqrt{\log n}} \geq \frac{2}{3\sqrt{3}}.$$

Keeping the same copies instead of deleting from them gives lower bounds for the complete pair-sum profile. The three statistics below live on different scales, but all three constants come from optimizing the same one-parameter construction.

Theorem 1.3 (Pair-sum multiplicities)**.** *We have*

$$\liminf_{n\to\infty} \frac{C(n)}{n^2\log n} \geq \frac{1}{8}, \qquad \liminf_{n\to\infty} \frac{D(n)}{n^{3/2}\sqrt{\log n}} \geq \frac{4}{3\sqrt{3}},$$

and

$$\liminf_{n\to\infty} \frac{\mu(n)}{\sqrt{n\log n}} \geq \frac{1}{2}.$$

Lower bounds arise naturally from explicit configurations; upper bounds require a different idea. A sum with many representations is centrally symmetric, and that symmetry can be converted into a large shifted product set. Combining this observation with a theorem of Jones and Roche-Newton [4] gives a complementary constraint.

Theorem 1.4 (Product-growth upper constraint)**.** *There exist absolute constants $c, C > 0$ such that*

$$M(2n) \geq c\,\frac{\mu(n)^{24/19}}{\big(\log(2+\mu(n))\big)^C}$$

for every $n \geq 1$. Consequently,

$$\mu(n) \leq M(2n)^{19/24+o(1)}$$

as $n \to \infty$.

This is not presently known to improve the elementary bound $\mu(n) \le \lceil M(n)/2 \rceil$, because sufficiently sharp control of $M(2n)$ in terms of $M(n)$ is not available. Its role is structural: it shows that unusually strong additive concentration inside one multinomial row forces multiplicative growth in the next doubled order.

The progression construction contains more information than the three lower limits above. Its exact pair-sum profile gives a lower envelope for $T(n,k)$ when k itself grows on the scale $\sqrt{n \log n}$, and it yields a corresponding additive-energy bound. We also examine two finite-dimensional questions. Near the top of the variable range, the last two variables create no new pair-sum multiplicities. At the other extreme, genuinely trinomial identities already form an explicit family with linearly many distinct colliding sums. These results help distinguish collisions inherited from the product embedding from collisions intrinsic to low-dimensional multinomial structure.

The paper follows the two uses of the product embedding. Section 2 proves the embedding and illustrates how additive relations lift. Sections 3–5 develop collision-free constructions and the general Sidon lower bound. Section 6 builds long progressions and separated copies; Section 7 uses their disjoint element ranges to force deletions from every Sidon subset. Section 8 begins the collision-rich half and develops general links among Sidon defects, rich sums, and product growth. Sections 9–11 use the disjoint pair-sum ranges of the same progression copies to obtain multiplicity and energy bounds. Sections 12 and 13 study variable-number stabilization and intrinsic trinomial collisions. The final section records exact data, computational details, OEIS connections, and open problems.

2 Multinomial coefficient sets and the product embedding

We keep the notation $\mathcal{M}_n$ from the introduction and write

$$M(n) = |\mathcal{M}_n|. \tag{1}$$

The function $M(n)$ was studied by Andrews, Knopfmacher, and Zimmermann [1]; in particular, they proved that $M(n) = o(p(n))$, where $p(n)$ is the ordinary partition function. The feature of $\mathcal{M}_n$ that drives the present paper is the following product embedding.

For finite sets $A_1, \ldots, A_\ell$ of positive integers, write

$$A_1 \cdots A_\ell = \{a_1 \cdots a_\ell : a_i \in A_i\}.$$

The embedding itself is elementary, but its usefulness comes from the freedom to choose the lower-order factors independently.

Proposition 2.1 (Product embedding)**.** *Let $n_1, \ldots, n_\ell$ be positive integers satisfying*

$$n_1 + \cdots + n_\ell = n.$$

Then

$$\binom{n}{n_1, \ldots, n_\ell} \mathcal{M}_{n_1} \cdots \mathcal{M}_{n_\ell} \subseteq \mathcal{M}_n.$$

Proof. For each i, choose a partition

$$p_{i,1} + \cdots + p_{i,t_i} = n_i$$

representing an element

$$x_i = \frac{n_i!}{p_{i,1}! \cdots p_{i,t_i}!} \in \mathcal{M}_{n_i}.$$

Concatenating these partitions gives a partition of n. The corresponding multinomial coefficient is

$$\frac{n!}{\prod_{i=1}^{\ell}\prod_{j=1}^{t_i} p_{i,j}!} = \binom{n}{n_1, \ldots, n_\ell} x_1 \cdots x_\ell,$$

which proves the inclusion. $\square$

In particular, if $r + m = n$, then

$$\binom{n}{r}\mathcal{M}_r\mathcal{M}_m \subseteq \mathcal{M}_n.$$

Since $1 \in \mathcal{M}_m$, this also gives

$$\binom{n}{r}\mathcal{M}_r \subseteq \mathcal{M}_n.$$

The special case $m = 1$ will be used repeatedly.

Corollary 2.2. *For every $n \geq 1$,*

$$(n+1)\mathcal{M}_n \subseteq \mathcal{M}_{n+1}.$$

Proof. Apply Proposition 2.1 with $n_1 = n$ and $n_2 = 1$, noting that $\mathcal{M}_1 = \{1\}$. $\square$

Consequently, every additive relation among elements of $\mathcal{M}_n$ may be lifted to $\mathcal{M}_{n+1}$ by multiplication by $n + 1$. For example, the collision $10 + 30 = 20 + 20$ in $\mathcal{M}_5$ becomes

$$60 + 180 = 120 + 120$$

in $\mathcal{M}_6$. More generally, the product embedding allows additive configurations in lower-order multinomial coefficient sets to be transferred, combined, and separated inside sets of higher order. The two principal applications developed below are the construction of large Sidon subsets and the construction of many disjoint families of pair-sum collisions.

3 Sidon subsets: elementary lifting

We begin with the simplest consequence of the product embedding. It gives a transparent linear lifting rule for Sidon subsets and, besides being useful for the small exact values, shows how additive structure at one order persists at every larger order.

Lemma 3.1. *Let S be a Sidon set of positive integers, and let $q \geq 3$. Then*

$$\{1\} \cup qS$$

is Sidon.

Proof. The pair sums are of the three forms

$$1 + 1, \qquad 1 + qx, \qquad qx + qy.$$

Their residues modulo q are respectively $2, 1, 0$, so sums of different forms cannot coincide. Equalities within the second form are trivial, while an equality within the third form reduces, after division by q, to an equality of pair sums in S. $\square$

Proposition 3.2. *For $2 \le r \le n$,*

$$s(n) \ge s(r) + n - r.$$

Consequently, $s(n) - n$ is nondecreasing for $n \ge 2$.

Proof. If $S \subseteq \mathcal{M}_k$ is Sidon, Proposition 2.1 and Lemma 3.1 show that

$$\{1\} \cup (k+1)S \subseteq \mathcal{M}_{k+1}$$

is Sidon. Iterating from $k = r$ to $k = n - 1$ proves the result. □

The iterated construction is explicit.

Proposition 3.3. *Let $2 \le r \le n$, and let $S \subseteq \mathcal{M}_r$ be Sidon. Then*

$$L_{n,r}(S) = \left\{1, n, n(n-1), \ldots, \frac{n!}{(r+1)!}\right\} \cup \frac{n!}{r!} S$$

is a Sidon subset of $\mathcal{M}_n$, of cardinality $|S| + n - r$.

Proof. This is the result of iterating $U \mapsto \{1\} \cup (k+1)U$ for $k = r, \ldots, n-1$. □

A simple special case gives a superincreasing family.

Corollary 3.4. *For every $n \ge 1$,*

$$\{1, n, n(n-1), \ldots, n!\}$$

is a Sidon subset of $\mathcal{M}_n$. In particular, $s(n) \ge n$.

Proof. Write

$$f_k = \frac{n!}{(n-k)!}, \qquad 0 \le k \le n-1.$$

Each f_k belongs to $\mathcal{M}_n$, corresponding to the partition $(n-k, 1, \ldots, 1)$. Moreover,

$$\frac{f_k}{f_{k-1}} = n - k + 1 \ge 2.$$

Suppose that $f_i + f_j = f_u + f_v$, where $i \le j$, $u \le v$, and, without loss of generality, $j \le v$. If $j < v$, then

$$f_i + f_j \le 2f_{v-1} \le f_v < f_u + f_v,$$

a contradiction. Hence $j = v$, and cancellation gives $f_i = f_u$, so $i = u$. □

The exact value $s(16) = 83$, proved computationally in Section 14, gives the concrete global estimate

$$s(n) \ge n + 67, \qquad n \ge 16.$$

This linear lifting is explicit and useful for the small exact values. The next construction shows that the product embedding can also amplify a seed set into many separated copies. Although the resulting explicit asymptotic bound is weaker than the general extraction bound in Theorem 1.1, it exposes a structural feature of the multinomial coefficient sets that will have an analogue on the collision-rich side.

4 Strongly Sidon sets and separated dilations

Ordinary Sidonicity is not quite enough when several scaled copies are superimposed. At the largest active scale, a contribution on one side of an equality may have to be compared with zero, one, or two contributions on the other side. The correct seed condition must therefore distinguish all sums made from zero, one, or two elements. This is exactly what the next definition records.

Definition 4.1. A finite set S of positive integers is *strongly Sidon* if all sums containing zero, one, or two elements of S are distinct, with repetition allowed in a two-element sum.

Equivalently, S is strongly Sidon if $\{0\} \cup S$ is Sidon. Let

$$b(n) = \max\{|S| : S \subseteq \mathcal{M}_n,\ S \text{ is strongly Sidon}\}.$$

Clearly $s(n) \geq b(n)$.

Lemma 4.2 (Separated-dilation lemma). *Let S be strongly Sidon, let $L = \max S$, and let*

$$1 \leq y_1 < \cdots < y_h$$

satisfy

$$y_{i+1} > 2Ly_i, \qquad 1 \leq i < h.$$

Then the sets y_iS are pairwise disjoint, and

$$U = \bigcup_{i=1}^{h} y_iS$$

is strongly Sidon. In particular, $|U| = h|S|$.

Proof. Suppose that two sums, each containing zero, one, or two elements of U, are equal. Let j be the largest index of a scale occurring in either sum. The contribution at scale y_j is y_jd, where

$$d \in D := \{0\} \cup S \cup (S + S).$$

Strong Sidonicity says that every element of D has a unique representation using zero, one, or two elements of S.

If $j = 1$, there is no contribution from a smaller scale. If $j > 1$, the contribution from all smaller scales on either side is between 0 and $2Ly_{j-1}$, so the difference between the two smaller-scale contributions has absolute value less than y_j. If the two coefficients d at scale y_j were different, their contributions would differ by at least y_j, which is impossible. Thus the coefficients are equal. Strong Sidonicity identifies the same multiset of zero, one, or two elements of S at scale y_j. Cancelling these terms and descending through the remaining scales proves uniqueness. In particular, one-element representations are unique, so the sets y_iS are pairwise disjoint. □

Proposition 4.3 (Separated-copy recursion). *Let $1 \leq r < n$, and let $S \subseteq \mathcal{M}_r$ be strongly Sidon with $L = \max S$. If $n \geq r + 2L$, then*

$$b(n) \geq |S|(n - r - 2L + 1).$$

Proof. Put $m = n - r$ and define

$$y_j = \frac{m!}{(m-j)!}, \qquad 0 \le j \le m - 2L.$$

These values belong to $\mathcal{M}_m$. For $0 \le j < m - 2L$,

$$\frac{y_{j+1}}{y_j} = m - j \ge 2L + 1 > 2L.$$

Lemma 4.2 shows that $\bigcup_j y_j S$ is strongly Sidon and has cardinality $|S|(m - 2L + 1)$. Proposition 2.1 places its common dilate

$$\binom{n}{r} \left(\bigcup_j y_j S \right)$$

inside $\mathcal{M}_n$. □

Corollary 4.4. *For integers $1 \le r < n$ with $n \ge r + 2r!$,*

$$b(n) \ge b(r)(n - r - 2r! + 1).$$

Proof. Apply Proposition 4.3 to a largest strongly Sidon subset of $\mathcal{M}_r$, using the crude bound $\max S \le r!$. □

5 Large Sidon subsets

We record both an explicit construction coming from the product embedding and the stronger nonconstructive lower bound supplied by general Sidon extraction. Keeping the two mechanisms separate is useful: the first reflects the internal multiplicative structure of $\mathcal{M}_n$, while the second gives the correct strongest conclusion presently available from the known cardinality estimates for $\mathcal{M}_n$.

5.1 An explicit elementary bound

Lemma 5.1. *For every integer $r \ge 3$, the set*

$$T_r = \left\{ \frac{r!}{(r-k)!} : 0 \le k \le r - 2 \right\} = \left\{ 1, r, r(r-1), \ldots, \frac{r!}{2} \right\}$$

is strongly Sidon. In particular, $|T_r| = r - 1$ and $\max T_r = r!/2$.

Proof. Write

$$t_k = \frac{r!}{(r-k)!}, \qquad 0 \le k \le r - 2.$$

Then

$$\frac{t_k}{t_{k-1}} = r - k + 1 \ge 3$$

for $1 \le k \le r - 2$. Hence

$$\sum_{j=0}^{k-1} t_j \le t_k \sum_{h=1}^{k} 3^{-h} < \frac{t_k}{2},$$

so

$$t_k > 2\sum_{j<k} t_j.$$

Suppose that two sums containing zero, one, or two elements of T_r are equal. At the largest index k where their multiplicities differ, the contribution has absolute value at least t_k, whereas the total possible contribution of all smaller terms is at most $2\sum_{j<k} t_j < t_k$, a contradiction. □

Proposition 5.2. *For every $r \geq 3$ and every $n \geq r + r!$,*

$$s(n) \geq b(n) \geq (r-1)(n-r-r!+1).$$

Proof. Apply the separated-dilation construction with T_r. Since $\max T_r = r!/2$, Proposition 4.3 gives

$$b(n) \geq |T_r|(n-r-2\max T_r+1) = (r-1)(n-r-r!+1).$$

□

Corollary 5.3. *As $n \to \infty$,*

$$s(n) \geq (1-o(1))\frac{n\log n}{\log\log n}.$$

Proof. Let r be the largest integer satisfying $r! \leq n/\log n$. Stirling's formula and maximality give

$$r\log r \sim \log n, \qquad r \sim \frac{\log n}{\log\log n}.$$

Moreover, $r/n \to 0$ and $r!/n \leq 1/\log n \to 0$. Proposition 5.2 now gives

$$s(n) \geq (r-1)(n-r-r!+1) = (1-o(1))\frac{n\log n}{\log\log n}.$$

□

5.2 General extraction and prime partitions

The explicit seed T_r shows how the product embedding can amplify a concrete collision-free pattern, but it has only $r-1$ elements. For the strongest existence theorem we use a different chain of ideas. Prime partitions guarantee that $\mathcal{M}_r$ itself contains very many distinct values; a general extraction theorem finds a large Sidon subset inside any set of that size; and the lemma below loses only a fixed factor when one asks for the stronger seed condition needed by separated dilations. Thus the argument passes through

prime partitions $\Longrightarrow$ $M(r)$ large $\Longrightarrow$ Sidon extraction $\Longrightarrow$ strong Sidon extraction.

We record the last step first.

Lemma 5.4 (Strong-Sidon extraction)**.** *Every finite Sidon set A of positive integers contains a strongly Sidon subset of cardinality at least $|A|/3$.*

Proof. Let $I = (1/3, 2/3) \subset \mathbb{R}/\mathbb{Z}$. For $\theta \in [0, 1)$, put

$$T_\theta = \{a \in A : \{\theta a\} \in I\},$$

where $\{x\}$ denotes the fractional part of x. For each nonzero integer a, the map $\theta \mapsto \theta a$ (mod 1) preserves Lebesgue measure. Hence

$$\int_0^1 |T_\theta| \, d\theta = \frac{|A|}{3},$$

so some T_θ has at least $|A|/3$ elements.

The interval I is sum-free modulo 1: if $\alpha, \beta \in I$, then $\alpha + \beta \pmod 1 \notin I$. Consequently no $x, y, z \in T_\theta$ satisfy $x + y = z$, including the case $x = y$. Since $T_\theta \subseteq A$ remains Sidon, and for positive Sidon sets the only additional obstruction to strong Sidonicity is a relation $x + y = z$, the set T_θ is strongly Sidon. □

Corollary 5.5. *For every $n \geq 1$,*

$$\left\lceil \frac{s(n)}{3} \right\rceil \leq b(n) \leq s(n).$$

Consequently, for $1 \leq r < n$ and $n \geq r + 2r!$,

$$s(n) \geq \left\lceil \frac{s(r)}{3} \right\rceil (n - r - 2r! + 1).$$

Proof. Apply Lemma 5.4 to a largest Sidon subset of $\mathcal{M}_n$ to obtain the first inequality; the second is immediate from the definition of $b(n)$. The recursive estimate follows from $s(n) \geq b(n)$ and Corollary 4.4. □

Let $p_{\mathbb{P}}(r)$ denote the number of unrestricted partitions of r into prime parts.

Proposition 5.6. *We have*

$$\liminf_{r \to \infty} \frac{\log b(r)}{\sqrt{r/\log r}} \geq \frac{\pi}{\sqrt{3}}.$$

Proof. Let $H(m)$ be the minimum, over all m-element subsets $B \subseteq \mathbb{Z}$, of the maximum cardinality of a Sidon subset of B. Abbott proved that there is an absolute constant $c_0 > 0$ such that

$$H(m) \geq c_0 \sqrt{m}$$

for all sufficiently large m [2]. More recently, Bailleul and Riblet proved the sharper estimate

$$H(m) \geq \left(\frac{1}{3\sqrt{3}} + o(1) \right) \sqrt{m}$$

as $m \to \infty$ [3]. The argument below requires only the established order of magnitude $H(m) \gg \sqrt{m}$. Since $M(r) \geq r$ by Corollary 3.4, the set $\mathcal{M}_r$ therefore contains a Sidon subset A with

$$|A| \geq c_0 \sqrt{M(r)}.$$

Lemma 5.4 gives

$$b(r) \geq \frac{c_0}{3} \sqrt{M(r)}.$$

By Theorem 1 of Andrews, Knopfmacher, and Zimmermann [1],

$$M(r) \geq p_{\mathbb{P}}(r)$$

for every positive integer r. The classical asymptotic for unrestricted prime partitions is

$$\log p_{\mathbb{P}}(r) \sim \frac{2\pi}{\sqrt{3}}\sqrt{\frac{r}{\log r}}$$

[5, 10]. Hence

$$\log b(r) \geq \frac{1}{2}\log p_{\mathbb{P}}(r) + O(1) = \left(\frac{\pi}{\sqrt{3}} + o(1)\right)\sqrt{\frac{r}{\log r}},$$

which proves the proposition. □

Proof of Theorem 1.1. Proposition 5.6 gives

$$\liminf_{n\to\infty} \frac{\log b(n)}{\sqrt{n/\log n}} \geq \frac{\pi}{\sqrt{3}}.$$

Since $s(n) \geq b(n)$, the same lower bound holds with $s(n)$ in place of $b(n)$. The equivalent ε-form follows immediately. Finally,

$$\frac{\sqrt{n/\log n}}{\log n} \longrightarrow \infty,$$

so the stated bound implies $s(n)/n^K \to \infty$ for every fixed $K > 0$. □

6 Arithmetic progressions and separated copies

We now turn from collision-free seeds to a configuration that is deliberately rich in additive relations: an interval. The first task is to realize a dilate of $\{1, \dots, q\}$ inside a multinomial row. The construction is multiplicative. We prepare a partition containing enough copies of every prime $p \leq q$ that any integer $j \leq q$ can be encoded by replacing selected parts p with $(p-1, 1)$. Each such replacement divides the factorial denominator by p, and hence multiplies the multinomial coefficient by p. Factoring j then produces the multiplier j.

Once one progression has been created, the product embedding places many copies at rapidly increasing scales. This is the point where the two halves of the paper meet: the disjoint element ranges force independent omissions from Sidon subsets, while the disjoint pair-sum ranges later let us add collision profiles without overlap.

For an integer $q \geq 2$, define

$$N(q) = \sum_{\substack{p \leq q \\ p \text{ prime}}} p \left\lfloor \frac{\log q}{\log p} \right\rfloor. \tag{2}$$

Proposition 6.1 (Arithmetic-progression construction)**.** *For every integer $q \geq 2$, the set $\mathcal{M}_{N(q)}$ contains an arithmetic progression*

$$\lambda, 2\lambda, \dots, q\lambda$$

for some positive integer λ.

Proof. For every prime $p \leq q$, put

$$e_p = \left\lfloor \frac{\log q}{\log p} \right\rfloor = \max\{e \geq 1 : p^e \leq q\}.$$

Consider the partition of $N(q)$ containing exactly e_p parts equal to p for every prime $p \le q$. Define

$$\Delta_q = \prod_{p \le q} (p!)^{e_p}, \qquad \lambda = \frac{N(q)!}{\Delta_q}.$$

Let $1 \le j \le q$, and write

$$j = \prod_p p^{f_p}.$$

Since $p^{f_p} \le j \le q$, we have $f_p \le e_p$. For every prime divisor p of j, replace f_p of the parts equal to p by the two parts $p-1$ and 1. The sum of the parts remains unchanged, while the product of their factorials changes from Δ_q to

$$\frac{\Delta_q}{\prod_p p^{f_p}} = \frac{\Delta_q}{j}.$$

The resulting multinomial coefficient is therefore $j\lambda$. Hence

$$\lambda, 2\lambda, \ldots, q\lambda \in \mathcal{M}_{N(q)}.$$

□

The size of the order $N(q)$ is asymptotically determined by the first powers of the primes.

Lemma 6.2. *As $q \to \infty$,*

$$N(q) \sim \frac{q^2}{2 \log q}.$$

Proof. Since a prime p contributes once for each integer $j \ge 1$ satisfying $p^j \le q$, we may write

$$N(q) = \sum_{j \ge 1} \sum_{p \le q^{1/j}} p.$$

By the prime number theorem and partial summation,

$$\sum_{p \le x} p \sim \frac{x^2}{2 \log x}.$$

Thus the term $j = 1$ contributes

$$\frac{q^2}{2 \log q}(1 + o(1)).$$

For $j \ge 2$, only primes $p \le \sqrt{q}$ occur, and there are at most $\log_2 q$ nonempty outer sums. Consequently,

$$\sum_{j \ge 2} \sum_{p \le q^{1/j}} p \le (\log_2 q) \sum_{p \le \sqrt{q}} p = O(q),$$

which is negligible compared with $q^2/\log q$. □

We now place many separated copies of the progression from Proposition 6.1 inside a single higher-order set.

Proposition 6.3 (Separated progression copies)**.** *Let $q \geq 2$ and*

$$n \geq N(q) + q.$$

Put

$$r = N(q), \qquad m = n - r, \qquad h = m - q + 1 = n - N(q) - q + 1.$$

Then $\mathcal{M}_n$ contains h arithmetic progressions

$$P_j = a_j\{1, 2, \ldots, q\}, \qquad 0 \leq j \leq h - 1,$$

such that

$$a_{j+1} > qa_j \qquad (0 \leq j < h - 1).$$

Consequently, the intervals

$$[a_j, qa_j], \qquad 0 \leq j \leq h - 1,$$

are pairwise disjoint, and the pair-sum intervals

$$[2a_j, 2qa_j], \qquad 0 \leq j \leq h - 1,$$

are also pairwise disjoint.

Proof. Let

$$\lambda\{1, \ldots, q\} \subseteq \mathcal{M}_r$$

be the progression supplied by Proposition 6.1. For

$$0 \leq j \leq m - q,$$

put

$$y_j = \frac{m!}{(m-j)!}.$$

The value y_j belongs to $\mathcal{M}_m$, since it is represented by the partition

$$(m - j, 1^j)$$

of m. Moreover, for $0 \leq j < m - q$,

$$\frac{y_{j+1}}{y_j} = m - j \geq q + 1 > q.$$

By Proposition 2.1,

$$P_j = \binom{n}{r}\lambda y_j\{1, \ldots, q\} \subseteq \mathcal{M}_n.$$

Setting

$$a_j = \binom{n}{r}\lambda y_j$$

gives

$$\frac{a_{j+1}}{a_j} = \frac{y_{j+1}}{y_j} > q.$$

Hence $qa_j < a_{j+1}$, so the intervals $[a_j, qa_j]$ are pairwise disjoint. Multiplying this inequality by 2 shows that

$$2qa_j < 2a_{j+1},$$

and therefore the intervals $[2a_j, 2qa_j]$ are pairwise disjoint as well. □

The first disjointness statement allows restrictions on a Sidon subset to be applied independently inside each progression P_j. The second ensures that the internal pair-sum representation profiles of different progression copies occur on disjoint sum values and may therefore be added without overlap.

7 A lower bound for the Sidon defect

We first use the separated progressions in a destructive way. A Sidon subset cannot keep an arbitrary number of points from a copy of $\{1,\ldots,q\}$: after rescaling, it can keep at most as many as an ordinary Sidon subset of the interval itself. Because the progression copies have disjoint element ranges, the required deletions add independently from copy to copy.

Let $F(q)$ denote the maximum size of a Sidon subset of $\{1,\ldots,q\}$. It is classical that

$$F(q) \sim \sqrt{q};$$

see [7] and the references therein.

Proposition 7.1. *For every integer $q \geq 2$ and every $n \geq N(q)+q$,*

$$M(n) - s(n) \geq (n - N(q) - q + 1)(q - F(q)).$$

Proof. By Proposition 6.3, the set $\mathcal{M}_n$ contains

$$h = n - N(q) - q + 1$$

pairwise disjoint dilates $P_j = a_j\{1,\ldots,q\}$. A Sidon subset meets each P_j in at most $F(q)$ elements, because dilation preserves pair-sum equalities. Thus at least $q - F(q)$ elements must be omitted from each of the h progressions, which proves the claim. □

Proof of Theorem 1.2. Choose an integer $q = q(n)$ satisfying

$$q \sim c\sqrt{n\log n}, \qquad 0 < c < 1.$$

Then $\log q \sim \frac{1}{2}\log n$, and Lemma 6.2 gives

$$N(q) \sim \frac{c^2 n\log n}{2\log q} \sim c^2 n.$$

In particular, $N(q) + q < n$ for all sufficiently large n. Moreover, $F(q) = o(q)$. Proposition 7.1 gives

$$M(n) - s(n) \geq \big((1-c^2) + o(1)\big)n\big(c + o(1)\big)\sqrt{n\log n}.$$

Hence

$$\liminf_{n\to\infty} \frac{M(n) - s(n)}{n^{3/2}\sqrt{\log n}} \geq c(1-c^2).$$

The maximum over $0 < c < 1$ is attained at $c = 1/\sqrt{3}$, with value $2/(3\sqrt{3})$. □

This completes the first use of the separated progressions. There we treated each copy as an obstruction: a Sidon subset was forced to delete many of its elements. We now reverse the viewpoint and keep every element. The same copies become sources of collisions, and their disjoint pair-sum intervals allow those sources to be counted independently.

8 Pair-sum representation functions and product growth

It is convenient to formulate the basic identities for an arbitrary finite set A of positive integers. For an integer t, define

$$r_A(t) = \#\{(x, y) : x, y \in A,\ x \leq y,\ x + y = t\}.$$

Set

$$\begin{aligned} C(A) &= \sum_t \max(r_A(t) - 1, 0), \\ D(A) &= \#\{t : r_A(t) \geq 2\}, \\ \mu(A) &= \max_t r_A(t), \\ T(A, k) &= \#\{t : r_A(t) \geq k\}. \end{aligned}$$

Thus $C(n) = C(\mathcal{M}_n)$, and similarly for D, μ, and T.

Proposition 8.1. *For every finite set A,*

$$C(A) = \binom{|A| + 1}{2} - |A + A|.$$

Moreover,

$$T(A, 2) = D(A), \qquad C(A) = \sum_{k=2}^{\mu(A)} T(A, k),$$

and

$$\mu(A) = \max\{k : T(A, k) > 0\}.$$

The sequence $k \mapsto T(A, k)$ is weakly decreasing.

Proof. There are $\binom{|A|+1}{2}$ unordered pairs with repetition. A sum value with r representations contributes $r - 1$ to the difference between the number of pairs and the number of distinct sums, proving the first identity. It also contributes once to each of $T(A, 2), \ldots, T(A, r)$, proving the second identity. The remaining assertions follow directly from the definitions. $\square$

Proposition 8.2. *Let A be a finite set, and let $s(A)$ denote the largest cardinality of a Sidon subset of A. Then*

$$\mu(A) \leq \min\left\{|A| - s(A) + 1, \left\lceil \frac{|A|}{2} \right\rceil\right\}.$$

In particular, the Sidon defect satisfies

$$|A| - s(A) \geq \mu(A) - 1.$$

Proof. Choose a sum t with $m = \mu(A)$ unordered representations. Distinct representations of the same sum have disjoint supports: if two shared an element, cancellation would make them the same unordered pair. Their union therefore has $2m$ elements, or $2m - 1$ if the diagonal representation $t/2 + t/2 = t$ occurs. This already gives $m \leq \lceil |A|/2 \rceil$.

If no representation is diagonal, a Sidon subset can contain both members of at most one of the m pairs, and at most one member of every other pair, so it contains at most $m + 1$ elements from their $2m$-element union. Hence

$$s(A) \leq |A| - 2m + (m + 1) = |A| - m + 1.$$

If the diagonal representation occurs, the union has $2m-1$ elements. A Sidon subset contains at most m of them, whether or not it contains $t/2$, and the same bound follows. $\square$

Proposition 8.3. *If $A \subseteq B$, then*

$$C(A) \le C(B), \qquad D(A) \le D(B), \qquad \mu(A) \le \mu(B), \qquad T(A,k) \le T(B,k).$$

Proof. Every representation by elements of A remains a representation by elements of B. Thus $r_B(t) \ge r_A(t)$ for every t, which implies all four inequalities. $\square$

By Corollary 2.2,

$$(n+1)\mathcal{M}_n \subseteq \mathcal{M}_{n+1}.$$

All four statistics are invariant under dilation by a positive integer. Proposition 8.3 therefore shows that $C(n)$, $D(n)$, $\mu(n)$, and $T(n,k)$ are weakly increasing in n for every fixed k.

We next turn to the upper constraint announced in the introduction. The reduction below is the bridge to the cited shifted-product theorem: all elements participating in one fixed rich sum form a centrally symmetric set, from which we extract a shifted product set.

Proposition 8.4. *There exist absolute constants $c, C > 0$ such that every finite set A of positive real numbers satisfies*

$$|AA| \ge c\,\frac{\mu(A)^{24/19}}{\big(\log(2+\mu(A))\big)^C}.$$

In particular,

$$|AA| \ge \mu(A)^{24/19-o(1)}$$

as $\mu(A) \to \infty$.

Proof. Choose t with $m = \mu(A)$ unordered representations. Let

$$B = \{x \in A : t - x \in A\}.$$

Then $B = t - B$, and its orbits under the involution $x \mapsto t-x$ are precisely the m unordered representations of t. Choose one element from each two-element orbit and, if the fixed point $t/2$ occurs, choose it as well. Let X be the set of chosen representatives. Then

$$|X| = m, \qquad X \subseteq A, \qquad t - X \subseteq A.$$

Hence

$$X(t-X) \subseteq AA.$$

Put $Y = -X/t$. Since the elements of A are positive and every $x \in X$ satisfies $0 < x < t$, we have $Y \subset (-1, 0)$. Moreover,

$$Y(Y+1) = -\frac{1}{t^2} X(t-X),$$

so these two product sets have the same cardinality. In the notation of Jones and Roche-Newton, Theorem 2 of [4] states that every finite $Y \subset \mathbb{R}$ satisfies

$$|Y(Y+1)| \gtrsim |Y|^{24/19}.$$

Their notation $U \lesssim V$ means $U \ll (\log V)^C V$ for some absolute C. Thus the preceding display gives

$$|Y|^{24/19} \ll \big(\log|Y(Y+1)|\big)^C |Y(Y+1)|.$$

Since trivially $|Y(Y+1)| \leq |Y|^2$, after harmless adjustment for small sets there are absolute constants $c, C > 0$ such that

$$|Y(Y+1)| \geq c \frac{|Y|^{24/19}}{\big(\log(2+|Y|)\big)^C}.$$

Therefore

$$|AA| \geq |X(t-X)| \geq c \frac{m^{24/19}}{\big(\log(2+m)\big)^C},$$

which proves the stated bound. □

Proof of Theorem 1.4. Apply Proposition 8.4 to $A = \mathcal{M}_n$. Proposition 2.1 with $n_1 = n_2 = n$ gives

$$\binom{2n}{n} \mathcal{M}_n \mathcal{M}_n \subseteq \mathcal{M}_{2n},$$

and therefore

$$|\mathcal{M}_n \mathcal{M}_n| \leq M(2n).$$

The first assertion follows immediately from Proposition 8.4. Since Theorem 1.3 implies $\mu(n) \to \infty$, the logarithmic factor is $\mu(n)^{o(1)}$, and hence

$$M(2n) \geq \mu(n)^{24/19 - o(1)}.$$

Equivalently,

$$\mu(n) \leq M(2n)^{19/24 + o(1)},$$

as claimed. □

9 The pair-sum profile of an arithmetic progression

The model object is the interval $I_q = \{1, 2, \ldots, q\}$. Its pair-sum profile has a simple triangular shape: as the target sum moves from 2 toward the center, new representations appear steadily; after the center, the pattern reverses. The calculation is elementary, but here the entire triangle matters because the separated copies from Section 6 have disjoint pair-sum intervals. Their internal profiles can therefore be transferred to $\mathcal{M}_n$ and added without overlap. Scaling does not change representation multiplicities.

Lemma 9.1. *For $2 \leq t \leq 2q$, let*

$$\rho_q(t) = \#\{(i,j) : 1 \leq i \leq j \leq q,\ i + j = t\}.$$

Then

$$\rho_q(t) = \begin{cases} \lfloor t/2 \rfloor, & 2 \leq t \leq q+1, \\ \lfloor t/2 \rfloor - t + q + 1, & q+1 < t \leq 2q. \end{cases}$$

For $2 \leq k \leq \lceil q/2 \rceil$, the values of t satisfying $\rho_q(t) \geq k$ are precisely

$$2k, 2k+1, \ldots, 2q - 2k + 2.$$

Thus

$$T(I_q, k) = 2q - 4k + 3.$$

Proof. For a fixed t, the first coordinate satisfies

$$\max(1, t-q) \le i \le \left\lfloor \frac{t}{2} \right\rfloor .$$

Counting these integers gives the formula for ρ_q. The inequality $\rho_q(t) \ge k$ is equivalent to $t \ge 2k$ on the increasing half and to $t \le 2q - 2k + 2$ on the decreasing half. The resulting interval contains $2q - 4k + 3$ integers. □

Corollary 9.2. *For $q \ge 3$,*

$$C(I_q) = \frac{(q-1)(q-2)}{2}, \qquad D(I_q) = 2q - 5, \qquad \mu(I_q) = \left\lceil \frac{q}{2} \right\rceil .$$

Proof. There are $\binom{q+1}{2}$ unordered pairs and $2q-1$ distinct sums, so

$$C(I_q) = \binom{q+1}{2} - (2q-1) = \frac{(q-1)(q-2)}{2}.$$

The identity $D(I_q) = T(I_q, 2) = 2q - 5$ follows from Lemma 9.1. The maximum of ρ_q equals $\lceil q/2 \rceil$. □

We also use the ordered additive energy

$$E^+(A) = \#\{(x_1, x_2, x_3, x_4) \in A^4 : x_1 + x_2 = x_3 + x_4\} = \sum_t R_A(t)^2,$$

where $R_A(t)$ counts ordered pairs $(x, y) \in A^2$ with $x + y = t$; see [9].

Lemma 9.3. *For every $q \ge 1$,*

$$E^+(I_q) = \frac{2q^3 + q}{3}.$$

Proof. The ordered representation counts are

$$1, 2, \ldots, q-1, q, q-1, \ldots, 2, 1.$$

Therefore

$$E^+(I_q) = q^2 + 2\sum_{j=1}^{q-1} j^2 = \frac{2q^3 + q}{3}.$$

□

10 Separated progressions and multiplicity lower bounds

We now combine the two ingredients: the exact profile of one interval and the existence of many separated copies inside $\mathcal{M}_n$. The important point is not merely that there are many copies, but that their pair-sum intervals are disjoint, so every copy contributes new sum values to the global profile.

Theorem 10.1. *Let $q \geq 3$ and $n \geq N(q) + q$, and put*

$$h = n - N(q) - q + 1.$$

Then

$$\begin{aligned} C(n) &\geq h\frac{(q-1)(q-2)}{2}, \\ D(n) &\geq h(2q-5), \\ T(n,k) &\geq h(2q-4k+3) \end{aligned}$$

for every $2 \leq k \leq \lceil q/2 \rceil$. Moreover,

$$\mu(n) \geq \left\lceil \frac{q}{2} \right\rceil.$$

Proof. Let $P_0, \ldots, P_{h-1}$ be supplied by Proposition 6.3. Their internal pair-sum intervals are pairwise disjoint, and each P_j is a dilation of I_q.

For every internal sum of P_j, all representations inside P_j remain representations in $\mathcal{M}_n$. Other elements of $\mathcal{M}_n$ can only increase its multiplicity. Since different copies concern distinct sum values, their contributions may be added. Lemma 9.1 and Corollary 9.2 give

$$C(n) \geq h\frac{(q-1)(q-2)}{2}, \qquad D(n) \geq h(2q-5),$$

and

$$T(n,k) \geq h(2q-4k+3)$$

for $2 \leq k \leq \lceil q/2 \rceil$. One copy already contains a sum with $\lceil q/2 \rceil$ representations, proving the assertion for $\mu(n)$. □

Proposition 10.2. *For $q \geq 2$ and $n \geq N(q) + q$,*

$$E^+(\mathcal{M}_n) \geq \big(n - N(q) - q + 1\big)\frac{2q^3 + q}{3}.$$

Proof. For each progression P_j, sum the squared ordered representation counts over its internal sum interval. These intervals are pairwise disjoint, so no sum value receives contributions from two different progression copies. The full representation count in $\mathcal{M}_n$ is at least the internal count of P_j. Lemma 9.3 gives the result. □

11 Asymptotic consequences for pair sums

There is a simple trade-off behind all three constants in Theorem 1.3. A longer progression creates more collisions inside each copy, but it also costs more of the available order n to manufacture the initial progression, leaving room for fewer separated copies. Lemma 6.2 shows that these two effects balance when the progression length is on the scale

$$q \asymp \sqrt{n \log n}$$

and optimize only the constant of proportionality. For fixed $0 < c < 1$, choose the integer

$$q = \left\lfloor c\sqrt{n \log n} \right\rfloor.$$

Then $q \sim c\sqrt{n \log n}$ and $q = o(n)$. Since $\log q \sim \frac{1}{2} \log n$, Lemma 6.2 gives

$$N(q) \sim \frac{q^2}{2 \log q} = \big(c^2 + o(1)\big)n, \qquad n - N(q) - q + 1 = \big(1 - c^2 + o(1)\big)n.$$

Proof of Theorem 1.3. Theorem 10.1 gives

$$C(n) \geq \left(\frac{1}{2}c^2(1-c^2) + o(1)\right) n^2 \log n.$$

The coefficient is maximized at $c = 1/\sqrt{2}$, where it equals $1/8$.

Similarly,

$$D(n) \geq \big(2c(1-c^2) + o(1)\big)\, n^{3/2}\sqrt{\log n}.$$

The coefficient is maximized at $c = 1/\sqrt{3}$, where it equals $4/(3\sqrt{3})$.

For the maximum multiplicity, any fixed $0 < c < 1$ gives

$$\mu(n) \geq \left(\frac{c}{2} + o(1)\right)\sqrt{n \log n}.$$

Taking the supremum over fixed $c < 1$ proves the last assertion. □

The finite estimate also retains information when the multiplicity threshold grows with n. For $0 \leq \alpha < 1/2$, put

$$c_\alpha = \frac{2\alpha + \sqrt{4\alpha^2 + 3}}{3}, \qquad \Phi(\alpha) = 2(c_\alpha - 2\alpha)(1 - c_\alpha^2).$$

Theorem 11.1. *Let $k_n \geq 2$ be integers such that*

$$\frac{k_n}{\sqrt{n \log n}} \longrightarrow \alpha, \qquad 0 \leq \alpha < \frac{1}{2}.$$

Then

$$\liminf_{n\to\infty} \frac{T(n, k_n)}{n^{3/2}\sqrt{\log n}} \geq \Phi(\alpha).$$

In particular, $\Phi(0) = 4/(3\sqrt{3})$ and $\Phi(\alpha) \to 0$ as $\alpha \uparrow 1/2$.

Proof. Fix c with $2\alpha < c < 1$, and choose $q = \lfloor c\sqrt{n \log n}\rfloor$. The hypothesis on k_n implies $k_n \leq \lceil q/2 \rceil$ for all sufficiently large n. Theorem 10.1 yields

$$T(n, k_n) \geq \big((1-c^2) + o(1)\big)n\big((2c - 4\alpha) + o(1)\big)\sqrt{n \log n}.$$

Hence

$$\liminf_{n\to\infty} \frac{T(n, k_n)}{n^{3/2}\sqrt{\log n}} \geq 2(c - 2\alpha)(1 - c^2).$$

The derivative of $(c - 2\alpha)(1 - c^2)$ vanishes when

$$3c^2 - 4\alpha c - 1 = 0.$$

For $0 \leq \alpha < 1/2$, its unique root in $(2\alpha, 1)$ is

$$c = c_\alpha = \frac{2\alpha + \sqrt{4\alpha^2 + 3}}{3}.$$

Substitution gives $\Phi(\alpha)$. □

Corollary 11.2. *We have*

$$\liminf_{n\to\infty} \frac{E^+(\mathcal{M}_n)}{n^{5/2}(\log n)^{3/2}} \geq \frac{4\sqrt{15}}{125}.$$

Proof. Proposition 10.2 gives

$$E^+(\mathcal{M}_n) \geq \left(\frac{2}{3}c^3(1-c^2) + o(1)\right) n^{5/2}(\log n)^{3/2}.$$

The coefficient is maximized at $c = \sqrt{3/5}$, where it equals $4\sqrt{15}/125$. □

12 Dependence on the number of variables

So far $\mathcal{M}_n$ has contained values coming from partitions with any number of parts. A natural question is how many variables are actually needed before the pair-sum statistics of the full set appear. Near the top of the variable range the answer is unexpectedly rigid: the last two adjunctions create no new repeated sums.

For $1 \le v \le n$, let

$$\mathcal{M}_{n,v} = \left\{ \frac{n!}{e_1! \cdots e_v!} : e_1 + \cdots + e_v = n, \quad e_i \ge 0 \right\}$$

with repeated values removed. Equivalently, $\mathcal{M}_{n,v}$ consists of the values corresponding to partitions of n into at most v positive parts. Thus

$$\mathcal{M}_{n,1} \subseteq \mathcal{M}_{n,2} \subseteq \cdots \subseteq \mathcal{M}_{n,n} = \mathcal{M}_n.$$

Define $C(n,v)$, $D(n,v)$, $\mu(n,v)$, and $T(n,v,k)$ by replacing $\mathcal{M}_n$ with $\mathcal{M}_{n,v}$.

Proposition 12.1. *For fixed n, all four statistics are weakly increasing in v.*

Proof. This follows from set inclusion and Proposition 8.3. □

Lemma 12.2. *Let $(p_1, \ldots, p_r)$ be a partition of n into r positive parts. Then*

$$\prod_{i=1}^{r} p_i! \ge 2^{n-r},$$

with equality if and only if every part is 1 or 2. Consequently, if $d \ge 0$ and $n \ge 2d$, then

$$\max \mathcal{M}_{n,n-d} = \frac{n!}{2^d},$$

achieved by the partition $(2^d, 1^{n-2d})$.

Proof. For every integer $a \ge 1$,

$$a! \ge 2^{a-1},$$

with equality exactly for $a = 1, 2$. Multiplication gives

$$\prod_{i=1}^{r} p_i! \ge 2^{\sum_i (p_i - 1)} = 2^{n-r}.$$

If a partition has at most $n-d$ parts, then $n - r \ge d$, so its factorial denominator is at least 2^d. When $n \ge 2d$, the displayed partition has exactly $n-d$ parts and denominator 2^d. □

In particular,

$$\max \mathcal{M}_{n,n-1} = \frac{n!}{2}, \qquad \max \mathcal{M}_{n,n-2} = \frac{n!}{4}, \qquad \max \mathcal{M}_{n,n-3} = \frac{n!}{8}$$

under the respective hypotheses $n \ge 2$, $n \ge 4$, and $n \ge 6$.

Proposition 12.3. *For every $n \geq 3$ and every $k \geq 2$,*

$$\begin{aligned} C(n, n-2) &= C(n, n-1) = C(n, n), \\ D(n, n-2) &= D(n, n-1) = D(n, n), \\ \mu(n, n-2) &= \mu(n, n-1) = \mu(n, n), \\ T(n, n-2, k) &= T(n, n-1, k) = T(n, n, k). \end{aligned}$$

For $n \geq 5$,

$$C(n, n-3) < C(n, n-2), \qquad D(n, n-3) < D(n, n-2).$$

Proof. The case $n = 3$ is immediate. Let $n \geq 4$. Passing from $\mathcal{M}_{n,n-2}$ to $\mathcal{M}_{n,n-1}$ adds exactly $n!/2$, corresponding to $(2, 1^{n-2})$. By Lemma 12.2, the largest old value is $n!/4$. Since both summands in an old pair are at most $n!/4$, every old pair sum is at most $n!/2$. Every sum involving the new value is at least $n!/2 + 1$, and the new sums are mutually distinct. No multiplicity at least 2 is created or altered.

Passing from $\mathcal{M}_{n,n-1}$ to $\mathcal{M}_{n,n}$ adds exactly $n!$, corresponding to (1^n). The largest previous value is $n!/2$, so, by the same two-summand bound, every old pair sum is at most $n!$. Every sum involving the new value is at least $n! + 1$, and the new sums are mutually distinct. This proves all four stabilization identities.

For $n \geq 6$, the values $n!/24$ and $n!/8$ already belong to $\mathcal{M}_{n,n-3}$, while $n!/6$ and $n!/4$ first occur in $\mathcal{M}_{n,n-2}$. Lemma 12.2 gives $\max \mathcal{M}_{n,n-3} = n!/8$, so every old pair sum is at most $n!/4$. The equality

$$\frac{n!}{4} + \frac{n!}{24} = \frac{n!}{6} + \frac{n!}{8} = \frac{7n!}{24}$$

is therefore a genuinely new colliding sum, proving strict increase of both C and D. For $n = 5$, use $10 + 30 = 20 + 20$. □

Remark 12.4. The stabilization is specific to the last two adjunctions. It does not assert stability in earlier columns, and the two-parameter growth of $C(n, v)$ and $D(n, v)$ remains open.

13 A family of trinomial collisions

The stabilization result above concerns the top of the variable range. At the opposite end, a different question arises: are trinomial collisions merely binomial identities lifted by adding a third part, or do genuinely three-variable families occur? They do. We now give an explicit intrinsic family that produces linearly many distinct colliding sums for every sufficiently large odd order.

Let $\mathcal{T}_n = \mathcal{M}_{n,3}$ and write $C_3(n) = C(\mathcal{T}_n)$ and $D_3(n) = D(\mathcal{T}_n)$. The family below is arranged so that, after division by a common factorial factor, the four multinomial coefficients collapse to four quadratic expressions. The collision then becomes an elementary polynomial identity; the remaining work is to prove that the resulting sums are genuinely distinct as the parameter varies.

For $m \geq 3$ and $2 \leq q \leq m$, define

$$A_q = \binom{2m+1}{m,\, m-q+1,\, q},$$
$$B_q = \binom{2m+1}{m+1,\, m-q+2,\, q-2},$$
$$U_q = \binom{2m+1}{m,\, m-q+2,\, q-1},$$
$$V_q = \binom{2m+1}{m+1,\, m-q,\, q}.$$

Lemma 13.1. *For every $m \geq 3$ and $2 \leq q \leq m$,*

$$A_q + B_q = U_q + V_q.$$

Proof. Divide all four terms by

$$K_q = \frac{(2m+1)!}{(m+1)!q!(m-q+2)!}.$$

The normalized terms are

$$A_q = (m+1)(m-q+2)K_q, \quad B_q = q(q-1)K_q,$$

$$U_q = (m+1)qK_q, \quad V_q = (m-q+2)(m-q+1)K_q.$$

The required equality reduces to

$$(m+1)(m-q+2) + q(q-1) = (m+1)q + (m-q+2)(m-q+1),$$

which follows by expansion. □

Let $S_q = A_q + B_q$. Direct simplification gives

$$S_q = \frac{\big((m+1)(m+2) - q(m+2-q)\big)(2m+1)!}{q!(m+1)!(m+2-q)!},$$

so $S_q = S_{m+2-q}$.

Lemma 13.2. *For distinct $q, r \in \{2, \dots, m\}$,*

$$S_q = S_r \iff q + r = m + 2.$$

Moreover, Lemma 13.1 gives a genuine collision except when m is even and $q = (m+2)/2$. The symmetric parameters q and $m+2-q$ describe the same equality with its sides reversed.

Proof. A direct calculation shows that the sign of $S_{q+1} - S_q$ is the sign of

$$(m+1-2q)\big(q^2 - qm - q + m^2 + 2m\big).$$

The second factor, viewed as a quadratic in q, has minimum

$$\frac{3m^2 + 6m - 1}{4} > 0.$$

Thus S_q is strictly increasing up to the center and strictly decreasing after the center. Together with $S_q = S_{m+2-q}$, this proves the first assertion.

Put $t = m - q + 2$. After division by K_q, the two unordered pairs are

$$\{(m+1)t,\ q(q-1)\} \qquad \text{and} \qquad \{(m+1)q,\ t(t-1)\}.$$

If corresponding entries agree, then $t = q$. Under crossed matching, division by the positive integers q, t would give both $t = m+2$ and $q = m+2$, impossible for $2 \le q, t \le m$. Hence the pairs coincide only when $t = q$, that is, when m is even and $q = (m+2)/2$.

Finally, for $q' = m + 2 - q$, comparison of factorial denominators gives

$$A_{q'} = U_q, \quad B_{q'} = V_q, \quad U_{q'} = A_q, \quad V_{q'} = B_q,$$

which proves the last statement. $\square$

Theorem 13.3. *For every odd $n = 2m + 1 \ge 7$,*

$$D_3(n) \ge \left\lfloor \frac{m-1}{2} \right\rfloor = \left\lfloor \frac{n-3}{4} \right\rfloor.$$

Consequently,

$$C_3(n) \ge \left\lfloor \frac{n-3}{4} \right\rfloor.$$

Proof. The involution $q \mapsto m + 2 - q$ acts on $\{2, \ldots, m\}$. By Lemmas 13.1 and 13.2, its nontrivial orbits give distinct genuine colliding sums. There are $\lfloor (m-1)/2 \rfloor$ such orbits. Each contributes at least one to the collision excess. $\square$

Every additive identity among binomial coefficients in row r lifts to a trinomial identity of every order $n \ge r$, since

$$\binom{n}{n-r, r-k, k} = \binom{n}{r}\binom{r}{k}.$$

Thus the trinomial problem receives both the intrinsic family above and lifted families from Pascal's triangle.

14 Exact computations, OEIS connections, and open problems

The asymptotic results describe what must eventually happen, while the first rows show how the additive structure actually emerges. The finite computations serve three purposes: they determine the first exact values of $s(n)$ and the collision statistics, they exhibit the first transitions from unique sums to double and then triple representations, and they independently check the finite identities used in the structural arguments above.

14.1 Exact Sidon-subset computations

For a finite set A, form a hypergraph $\mathcal{H}(A)$ whose vertex set is A, and whose hyperedges are the supports of nontrivial equalities

$$x + y = u + v$$

between distinct unordered pairs from A. A subset of A is Sidon exactly when it is independent in $\mathcal{H}(A)$. Consequently, $M(n) - s(n)$ is the transversal number of $\mathcal{H}(\mathcal{M}_n)$.

For each n, we generated the distinct values in $\mathcal{M}_n$, formed the inclusion-minimal supports of nontrivial pair-sum collisions, and solved the resulting minimum-transversal problem by SAT with a cardinality constraint. For $n = 16$, one has $|\mathcal{M}_{16}| = 157$; the reduced hypergraph contains 146 active vertices and 2581 minimal forbidden supports. A transversal of size 74 exists, while none of size 73 exists. Hence

$$s(16) = 157 - 74 = 83.$$

The exact values are

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$s(n)$	1	2	3	5	6	8	11	15	20	25	30	39	45	55	66	83

For $n = 5$,

$$\mathcal{M}_5 = \{1, 5, 10, 20, 30, 60, 120\}, \qquad 10 + 30 = 20 + 20.$$

The set

$$\{1, 5, 20, 30, 60, 120\}$$

is Sidon, which gives the elementary verification $s(5) = 6$.

14.2 Pair-sum data

The first values of the principal collision statistics are as follows.

n	$M(n)$	$C(n)$	$D(n)$	$\mu(n)$
1	1	0	0	1
2	2	0	0	1
3	3	0	0	1
4	5	0	0	1
5	7	1	1	2
6	11	6	6	2
7	14	8	8	2
8	20	24	24	2
9	27	43	42	3
10	36	74	68	3
11	47	148	128	4
12	64	245	198	4
13	79	441	336	5
14	102	753	557	6
15	125	1178	781	6
16	157	1742	1119	7
17	193	2605	1623	8
18	243	3805	2311	9
19	296	5589	3321	9
20	366	8256	4809	10

For example,

$$\mathcal{M}_6 = \{1, 6, 15, 20, 30, 60, 90, 120, 180, 360, 720\}.$$

There are $\binom{12}{2} = 66$ unordered pairs with repetition but only 60 distinct sums, so $C(6) = 6$. At $n = 9$, the sum 1764 has three representations,

$$84 + 1680 = 252 + 1512 = 504 + 1260,$$

which explains the first strict inequality $C(n) > D(n)$.

The sequences $s(n)$, $C(n)$, $D(n)$, and $\mu(n)$ are OEIS A398429, A398233, A398238, and A398263, respectively. The cumulative profile is recorded in A398265. The variable-number triangles $C(n, v)$ and $D(n, v)$ are A398232 and A398237, and the trinomial statistics $C_3(n)$ and $D_3(n)$ are A398230 and A398231 [8].

14.3 Scope of the computational checks

The accompanying Maple code constructs the collision hypergraphs and carries out the SAT computations for $s(n)$. The hypergraphs and SAT instances grow rapidly; $n = 16$ is the largest order for which the present exhaustive pipeline was run to a completed optimality result. A second program generates $\mathcal{M}_{n,v}$ and the pair-sum statistics, reproduces the table through $n = 20$, checks the stabilization statement and its preceding strict increase through $n = 20$, verifies the trinomial identities through $m = 80$, and checks the arithmetic-progression construction through $q = 10$. An independent standard-library Python implementation reproduces the pair-sum calculations. All finite calculations use exact integer arithmetic. The asymptotic results and the symbolic proofs of the constructions are independent of these checks. The SAT optimality calculation is reproducible from the supplied source and console output, but no solver-independent unsatisfiability certificate is included.

14.4 Open problems concerning Sidon subsets

The remaining questions on the collision-free side fall into three levels: asymptotic size, exact small values, and the relation between ordinary and strongly Sidon subsets.

Question 14.1. What is the order of magnitude of $s(n)$? Theorem 1.1 shows that $s(n)$ is superpolynomial in n. Is $\log s(n)$ of order $\sqrt{n/\log n}$, or does the additional structure of $\mathcal{M}_n$ force substantially faster growth?

Question 14.2. Determine $s(17)$ and further exact values. More generally, can the exact computation be pushed far enough to reveal stable finite-scale behavior not visible in the asymptotic bounds?

Question 14.3. What is the order of magnitude of the Sidon defect $M(n) - s(n)$? How does the ratio $b(n)/s(n)$ behave, where $b(n)$ is the largest size of a strongly Sidon subset?

14.5 Open problems concerning pair-sum multiplicities

On the collision-rich side, the main unknowns are the true global scales, the possible limiting shape of the multiplicity profile, and the classification of collisions that are intrinsic rather than inherited. The progression constructions in this paper are essentially one-sided: they produce lower bounds, while corresponding upper bounds appear to require different information about the arithmetic structure of $\mathcal{M}_n$.

Question 14.4. Determine the true orders of magnitude of $C(n)$, $D(n)$, $\mu(n)$, and $E^+(\mathcal{M}_n)$. Are the lower bounds proved here of the correct scale? Theorem 1.4 bounds

$\mu(n)$ in terms of $M(2n)$, but it is not known whether this improves the elementary estimate $\mu(n) \leq \lceil M(n)/2 \rceil$ from Proposition 8.2. Can one prove a genuine power saving

$$\mu(n) \leq M(n)^{1-\delta}$$

for some absolute $\delta > 0$?

Question 14.5. Does the normalized cumulative profile $k \mapsto T(n, k)$ approach a limiting shape on the scale $k \asymp \sqrt{n \log n}$? How close is the explicit function Φ to the true lower envelope?

Question 14.6. For fixed $v \geq 3$, determine the growth of the v-nomial collision statistics $C(n, v)$ and $D(n, v)$, and classify the primitive pair-sum identities that are not inherited from lower order by the product embedding.

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OpenAI ChatGPT was used during the exploration and preparation of this work to suggest and stress-test proof approaches, assist with symbolic manipulations and Maple/Python code, check computational workflows, and revise language. The author determined the research direction and final selection of results. AI output was not treated as evidence: every theorem, proof, and cited source included in the manuscript was checked by the author, and the computational results were run or independently cross-checked as described above. The author takes full responsibility for the mathematical claims, computations, and submitted text.

The author thanks the OEIS community for providing the setting in which the motivating sequences were developed.

Code and computational material

The ancillary archive contains the Maple sources for the Sidon and pair-sum computations, the completed SAT console output for $n = 14, 15, 16$, a computational supplement describing the SAT encoding and environment, an independent Python verification, recorded successful audit outputs, and a README with reproduction instructions. No external dataset is required.

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